

Indian Institute of Technology Kharagpur
Department of Mathematics
MA11003 - Advanced Calculus
Problem Sheet -13A

1. Evaluate the triple integral

(i) $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dz dy dx$ directly.

(ii) $\int_0^5 \int_0^{\sqrt{25-x^2}} \int_0^6 \frac{dz dy dx}{\sqrt{x^2+y^2}}$ using cylindrical coordinates.

(iii) $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dz dy dx}{1+x^2+y^2+z^2}$ using spherical coordinates.

2. Compute $\iiint xy \, dz dy dx$ over the volume enclosed by planes $z = x + y$ and $z = 0$, and between the surfaces $y = x^2$ and $x = y^2$.

3. Compute $\iiint \frac{z \, dz dy dx}{\sqrt{x^2+y^2}}$ if the region is bounded by paraboloid $x^2 + y^2 = 2z$ and the cylinder $x^2 + y^2 = 4$ in the first octant.

4. Find the integral $\iiint e^{(x^2+y^2+z^2)^{3/2}} dx dy dz$ in the region $R = x, y, z \geq 0, x^2 + y^2 + z^2 \leq 1$ using spherical coordinates.
